Assignment P4 – The Agulhas Region:

Introduction:

The Agulhas region contains a western boundary current that flows down and along the East coast of South Africa and retroflects at the base of the continent. It originates from the Eastern Madagascan current and the Mozambiquan current. It is a unique system, being the only western boundary current that flows between two subtropical gyres. It is a very significant current as it produces large eddies, known as Agulhas rings. These are nutrient-poor systems that are anticyclonic and flow in an anticlockwise direction.

I am fascinated with how the continental shelf attracts many animals towards the coast, especially during the annual Sardine Run. The retroflection and the Agulhas eddies are also significant systems that influence the entire surrounding ocean, transporting salt and oligotrophic waters, from the Tropical regions on the East of Africa into the Atlantic on the West. The Agulhas even feeds saline water into the Gulf Stream, emphasizing how necessary this current is for the Global ocean circulation pattern.

*Note: Chlorophyll a is referred to as “chlorophyll” or “chlorophyll-a” throughout this document.

Figures:

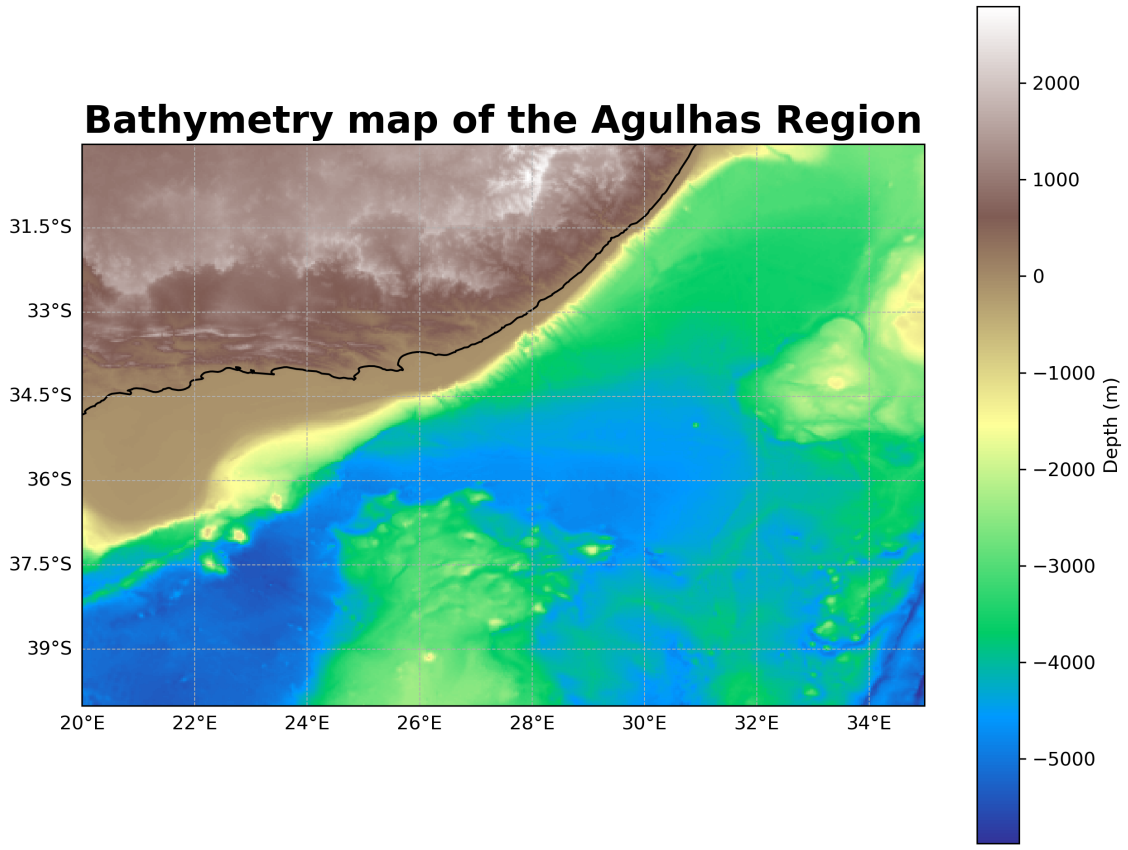

Figure 1: Bathymetry map of the Agulhas region. The continental shelf is visible on the left of the map, in orange. It sits at shallower than 1000m deep, as it runs down the coast and extends outwards from 34ºS to 37ºS within the 20ºE to 23ºE.

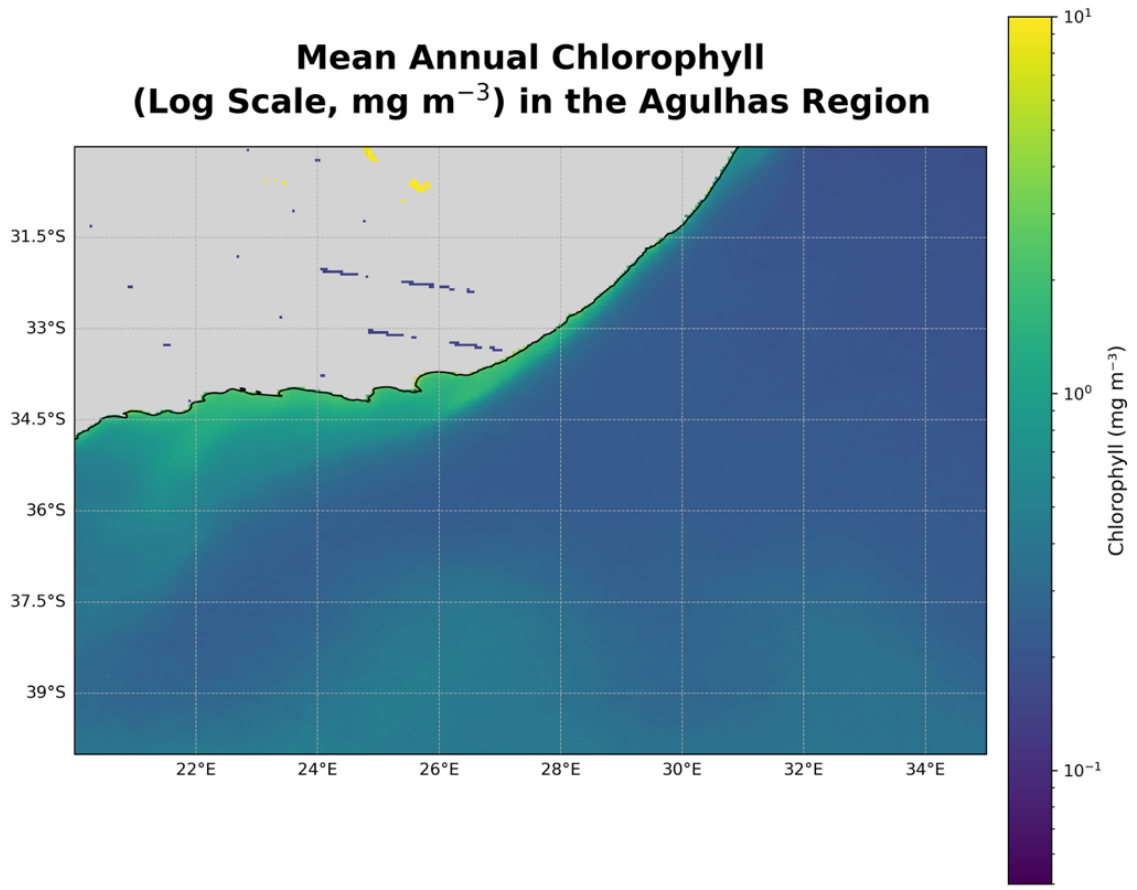

Figure 2: Map showing the chlorophyll-a (mg/m^3) concentrations on a log scale of the Agulhas region. The chlorophyll concentration is highest at the coast on the 34.5ºS line. The surrounding area has low chlorophyll concentrations.

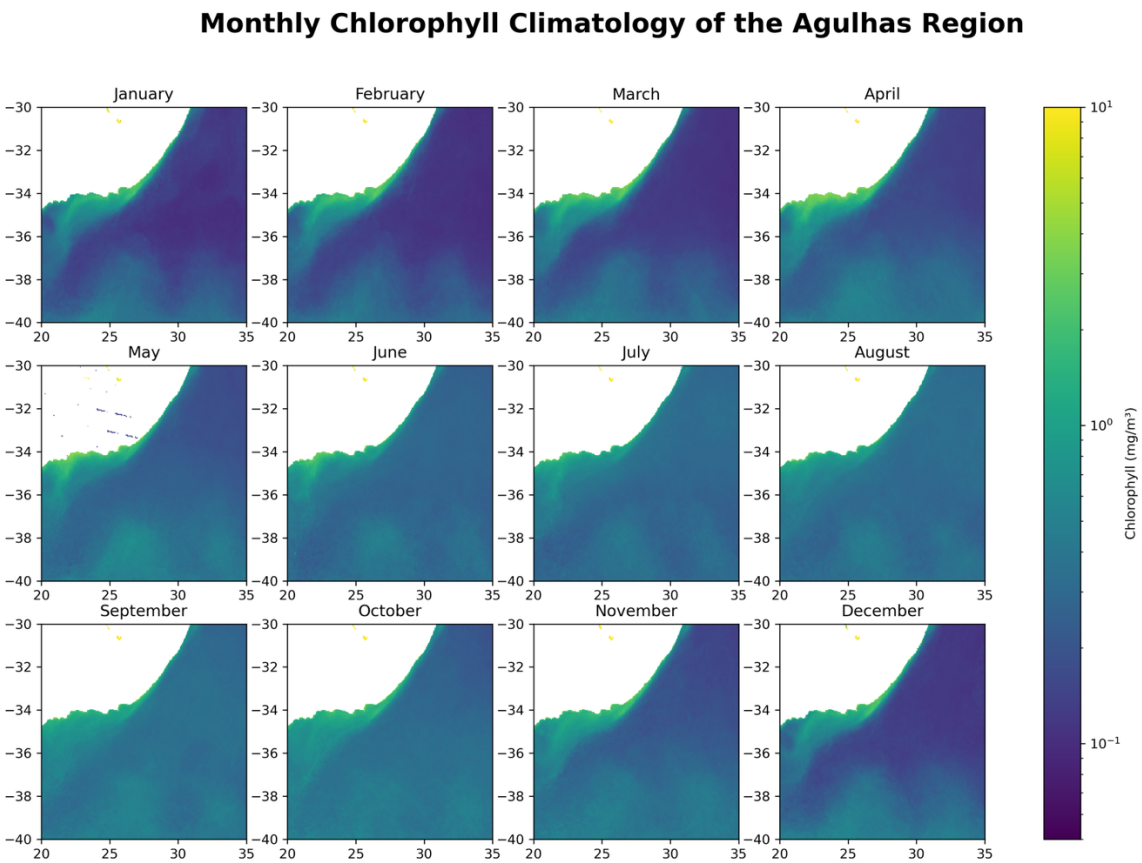

Figure 3: A plot showing the monthly mean concentrations of chlorophyll-a concentrations (mg/m^3) for each month of the year. Concentrations increase during the Autumn and winter months, starting in April and reaching a maximum in September and October. From then, concentrations decrease during Summer, with a low February and March.

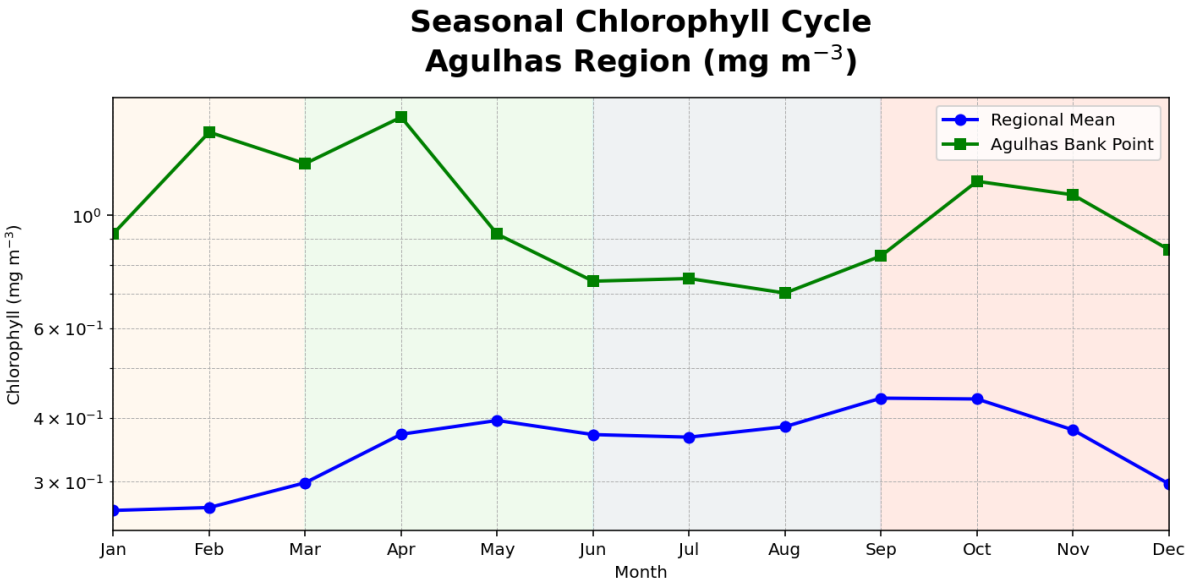

Figure 4: Time series of chlorophyll-a concentrations (mg m^{-3}) at 34.5°S, 23°E (the highest point on the Agulhas bank), compared with the regional mean concentration of chlorophyll over the annual cycle. The blue line is the regional mean, the concentrations increase during the colder Autumn and Winter and then decrease during the summer months. The chosen point has much higher chlorophyll concentrations that the regional mean, the point is situated at the coast, where productivity is at a maximum. It reaches a maximum concentration in April and a minimum in August. This varies from the regional trend, as the maximum is in September and minimum in January regionally.

Datasets

ESA-CCI Ocean Colour Climatology website was used to download the Chlorophyll-a data. ("Ocean Colour", n.d.) (Sathyendranath et al., 2019)

Version 6:
Sathyendranath, S.; Jackson, T.; Brockmann, C.; Brotas, V.; Calton, B.; Chuprin, A.; Clements, O.; Cipollini, P.; Danne, O.; Dingle, J.; Donlon, C.; Grant, M.; Groom, S.; Krasemann, H.; Lavender, S.; Mazeran, C.; Mélin, F.; Müller, D.; Steinmetz, F.; Valente, A.; Zühlke, M.; Feldman, G.; Franz, B.; Frouin, R.; Werdell, J.; Platt, T. (2023): ESA Ocean Colour Climate Change Initiative (Ocean_Colour_cci): Monthly climatology of global ocean colour data products at 4km resolution, Version 6.0. NERC EDS Centre for Environmental Data Analysis, 27 March 2026
https://catalogue.ceda.ac.uk/uuid/690fdf8f229c4d04a2aa68de67beb733/

Global Multi-Resolution Topography Data Synthesis (GMRT) map tool was used to create a grid file of the Agulhas region and a downloadable netCDF file. Used to plot bathymetry. ("GMRT MapTool", n.d.)

References:

GMRT MapTool. n.d. Available: https://www.gmrt.org/GMRTMapTool/ [2026, March 28].
Ocean Colour. n.d. Available: https://climate.esa.int/en/projects/ocean-colour/ [2026, March 28].
Sathyendranath, S., Brewin, R.J.W., Brockmann, C., Brotas, V., Calton, B., Chuprin, A., Cipollini, P., Couto, A.B., et al. 2019. An Ocean-Colour Time Series for Use in Climate Studies: The Experience of the Ocean-Colour Climate Change Initiative (OC-CCI). *Sensors*. 19(19):4285. DOI: 10.3390/s19194285.